With 750 labelled tweets, does fine-tuning beat a calibrated open LLM?

Research question (unchanged)

Take **750 labelled tweets**. Does fine-tuning **BERTurk** beat prompting an open LLM (**Gemma-2-9B-it** or **Qwen3-8B**) on **macro-F1** over the official OffensEval-TR 2020 test set (3,528 tweets), when the LLM gets the same 750 labels: 8 as prompt examples and the other 742 to set its decision threshold?

We call it a no when

against either LLM, BERTurk's mean over 10 runs is less than **0.02** above the LLM's mean over 3 runs, or a paired bootstrap test says the gap could be chance.

0.763
TF-IDF baseline on the frozen test set (95% CI 0.745–0.779)

60
baseline errors audited, grouped into categories with counts

0.790
Gemma-4-12B few-shot dev macro-F1 after threshold calibration

Vesile Han · Gürkan Göztepeli (CSE 654, research leads) · Mert Uğur · Utku Gökçek (CSE 484). 10 talk slides + 4 Project Card appendix slides.

30,000 train / 1,756 dev / 3,528 frozen test; 19.3% offensive

File / split	Tweets	NOT	OFF	OFF %
Official training file	31,756	25,625	6,131	19.3
Our train (seed 42)	30,000	24,208	5,792	19.3
Our dev	1,756	1,417	339	19.3
Official test (frozen)	3,528	2,812	716	20.3

OffensEval-TR 2020 (Çöltekin, LREC 2020). Annotations CC BY 2.0. @USER/URL already masked; no extra cleaning. 9 duplicate train texts; 1 train/test overlap.

REAL EXAMPLE · OFF · TRAIN ID 33518

“Kırmızı ağır karar Allah belanı versin”

“A red card is a harsh call, God damn you.” Swearing at a decision, not a person; still OFF.

CONTRAST · NOT

“…annem alacağı arabadan kaç kere vazgeçti ya kahretsin zalım trafik”

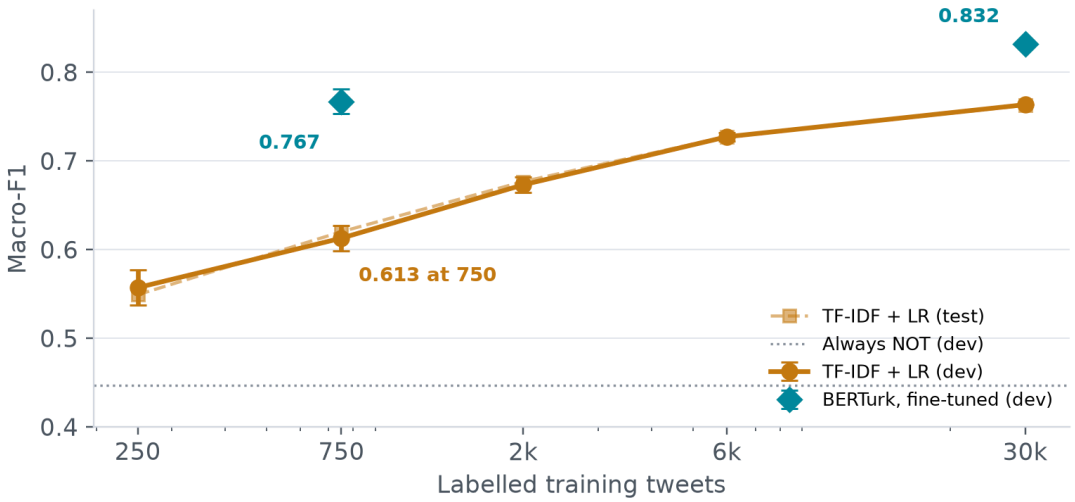
“How many times my mum gave up on buying a car, damn this cruel traffic.” Cursing at traffic is not offensive; target matters.

Why these counts matter. Only 19.3% of training tweets are offensive. If the model always answers “not offensive”, accuracy is high at 80.7%, but macro-F1 is 0.447. Accuracy alone hides the failure, which is why we use macro-F1. The test set stays frozen; dev is only for code checks and mid-project numbers, and every system uses the saved subsets (seed 42 + k).

TF-IDF + LR reaches 0.763 on test; 0.620 with 750 labels

System	Test macro-F1	OFF P	OFF R	Spread
Always “not offensive”	0.444	—	0.000	deterministic
TF-IDF + LR, 750	0.620 ± 0.012	—	—	sd over 10 samples
TF-IDF + LR, 30,000	0.763	0.589	0.673	95% CI [0.745, 0.779]

The frozen test set was scored once for this deterministic baseline. All method results stay on dev until the final freeze.



Solid: TF-IDF dev. Dashed: TF-IDF test. Diamonds: fine-tuned BERTurk dev.

Where the floor sits. With 750 labels, the simple baseline scores 0.620 on test and 0.613 on dev. That is the minimum BERTurk and the LLMs must beat. At 30,000 labels, TF-IDF climbs to 0.763; BERTurk scores 0.767 with only 750 labels and 0.832 with 30,000 on dev.

275 dev errors; a 60-error audit shows where a word model fails

False alarms (166) — category	n
No insult / spurious cue	9
Political or religious criticism	10
Sports criticism or banter	5
Personal criticism	3
Sarcasm or irony	1
Label noise / ambiguous	2

Missed offensive (109) — category	n
Implicit insult, no profanity	8
Masked or abbreviated profanity	5
Veiled threat or violence	3
Political or religious insult	4
Institutional or sports criticism	3
Label noise / not offensive	7

Counts are in the 60-error audit (30 per direction, seed 42); each table sums to 30.

FALSE ALARM · F3 · 13105
“Aziz Yıldırım Fenerbahçe’yi padişahlık zannediyor herhalde…”
Club criticism without a slur.

MISS · M1 · 43325
“…bundan daha parazit canlı türü yok”
Implicit insult with no swear word.

MISS · M2 · 15069
“…bakanlık seçimi kaldı amkk”
Masked profanity, unseen by the word/char model.

The review found that false alarms often involve conflict and club words and character combinations. Implicit insults and masked profanity are easily missed. About one in seven audited errors is label noise, so the ceiling is below 1.0.

Both arms run end-to-end; first dev numbers

Controlled change: the baseline trains its weights on the 750 labels. The LLM uses those same 750 labels to pick 8 examples and a threshold. Nothing else differs; only how the labels are used.

LLM arm

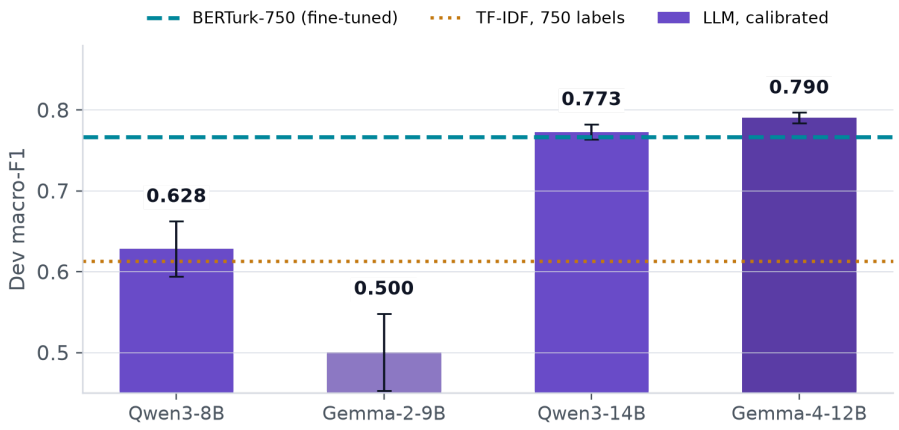
750 labels → 8 prompt examples (4 OFF / 4 NOT) → 742 labels set the threshold → score dev as log P(A) − log P(B).

Fine-tuning arm

BERTurk and XLM-R-base, same subsets: 128 tokens, batch 16, lr 2e-5, 4 epochs or 400 updates, no early stopping. 46 runs.

System (dev)	Macro-F1	Runs	Threshold
Qwen3-8B, zero-shot	0.162	1	0
Qwen3-8B, 8 + 742	0.628 ± 0.034	3	3.13 / 2.75 / 2.50
Gemma-2-9B, zero-shot	0.412	1	0
Gemma-2-9B, 8 + 742	0.500 ± 0.048	3	0.63 / 0.25 / 0.88
BERTurk, 750	0.767 ± 0.014	10	—
BERTurk, 30,000	0.832 ± 0.003	3	—

BERTurk beats both proposal LLMs; Gemma-4-12B leads on dev



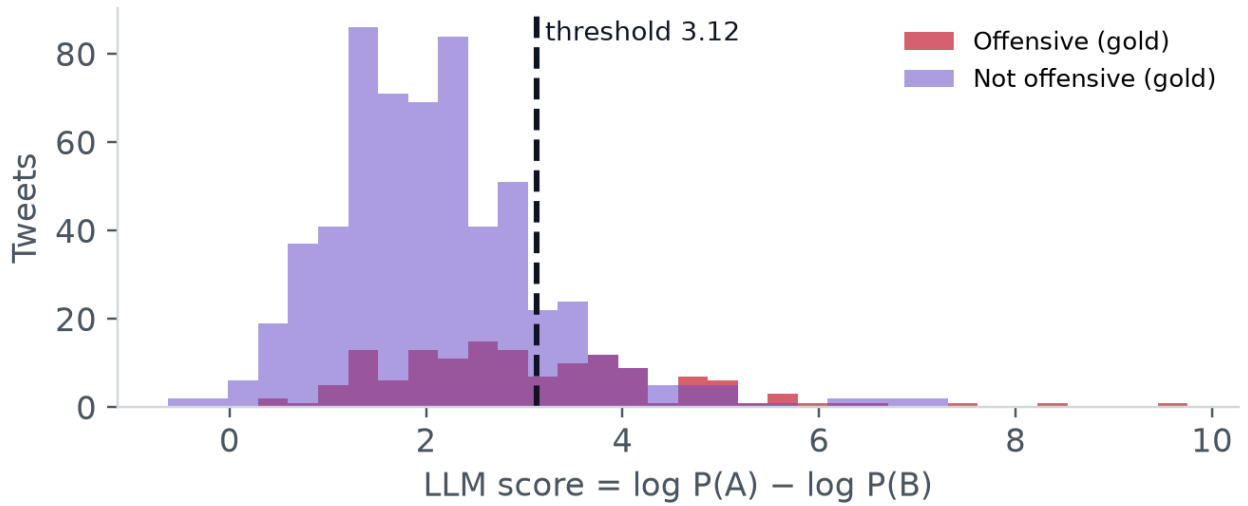
System	Dev macro-F1	Runs
TF-IDF + LR, 750	0.613 ± 0.015	10
Qwen3-8B	0.628 ± 0.034	3
Gemma-2-9B	0.500 ± 0.048	3
Qwen3-14B	0.773 ± 0.009	3
Gemma-4-12B	0.790 ± 0.007	3
BERTurk, 750	0.767 ± 0.014	10
BERTurk, 30,000	0.832 ± 0.003	3

BERTurk-750 outperforms TF-IDF by 0.154 and Qwen3-8B by 0.139; it is 0.267 ahead of Gemma-2-9B. Gemma-4-12B is 0.023 ahead of BERTurk-750. The test comparison is still pending.

Qwen is shifted; Gemma's threshold is already near optimal

Zero-shot diagnostic	F1 @ 0	F1 @ best	AUC	mean OFF	mean NOT
Qwen3-14B	0.171	0.677	0.756	2.79	2.10
Gemma-4-12B	0.763	0.765	0.864	1.03	-4.13

F1 @ 0 uses threshold 0; F1 @ best is the best threshold. Mean OFF/NOT are mean log P(A) − log P(B). Chart: Qwen3-8B, one run.



What explains the gap

- **It is not the tokenizer.** Gemma packs these tweets more tightly: 32.3 vs 37.9 tokens/tweet.
- **Qwen is shifted, not blind.** At threshold 0 it calls 99% of tweets offensive, but the ranking still works: best-threshold F1 is 0.677.
- **Gemma is already centred.** F1 at 0 is 0.763 and the best threshold reaches 0.765, so threshold tuning has almost nothing to gain here.
- **The 8 examples still help Qwen:** 0.677 → 0.773 after calibration.

This is threshold geometry, not proof of contamination. It does not separate pretraining language from alignment; the exact-completion check is still planned.

One dev pass on an L4: 3–5 tweets/s with examples, 10–18 without

System	Params	Dev scoring	Throughput	Peak VRAM	Notes
TF-IDF, 750	—	0.6 s	~3,100/s	CPU	fit 1.0 s
TF-IDF, 30,000	—	0.6 s	~3,100/s	CPU	fit 14.6 s
Qwen3-8B, few-shot	8.2B	355 s	5.0/s	—	3 runs
Gemma-2-9B, few-shot	9.2B	430 s	4.1/s	—	3 runs
Qwen3-14B, few-shot	14.8B	644 s	2.7/s	—	3 runs
Gemma-4-12B, few-shot	12.0B	579 s	3.0/s	—	3 runs
Qwen3-14B, zero-shot	14.8B	180 s	9.8/s	10.7 GB	load 157 s · 1,290 tok/s

Scoring the 1,756 dev tweets; model download and load are excluded unless stated. Few-shot is 3–4× slower because each prompt carries 8 examples (~600 tokens).

BERTurk: 35 s training at 750 labels, ~29 min at 30,000. Gemma-4-26B-A4B exceeded 22 GB and could not load on L4. Reported runs: 9.4 Colab units (~2.5 L4-hours).

BERTurk and Ablation A are done; remaining steps have a date and an owner

Date	Experiment	Owner	Status
11 Nov	BERTurk 750 (10) + 30,000 (3) on dev	Mert	done
11 Nov	Ablation A: BERTurk vs XLM-R, 750	Vesile	done
12 Nov	Mid-project talk; freeze this deck	All	talk
21–24 Nov	Ablation B: Gemma with English instructions	Gürkan	planned
25–27 Nov	Test-contamination check, first 200 test tweets	Gürkan	planned
28–30 Nov	Freeze methods; open test; final runs	All	planned
1–5 Dec	Paired bootstrap, error analysis, paper	Mert · Utku	planned
8 Dec	Final report	All	planned

Biggest risk

A 26B+ model may not fit the L4: Gemma-4-26B-A4B OOM'd at 22 GB, so we used the 12B dense model. **Fallback:** Qwen3-14B alone.

Result risk

The lines may never cross. The 750-label question still has a clear answer; the learning curve is descriptive.

On proposal feedback: the rubric asks for test-set baseline numbers, so we scored the deterministic TF-IDF baseline once on test and kept methods on dev.

What we know, what we do not, and what comes next

BASELINE · TEST

0.763

TF-IDF test macro-F1 at 30,000 labels. At 750 labels it falls to 0.620 ± 0.012. Deterministic; test 95% CI [0.745, 0.779].

BEST LLM · DEV

0.790

Gemma-4-12B dev macro-F1 after calibration; zero-shot 0.763. Qwen3-14B reaches 0.773. Threshold from 742 labels; 3 runs.

FINE-TUNE · DEV

0.767

BERTurk-750 dev (10 runs); 30,000 labels reaches 0.832. It beats Qwen3-8B and Gemma-2-9B, but Gemma-4-12B leads on dev. Gap to Gemma-4-12B: −0.023.

What we do not have yet: any test-set method result. BERTurk-750 is 0.767 on dev; the final answer uses test. Fixed before we ran anything: the test set, macro-F1, the 0.02 minimum gap, and the paired bootstrap.

Questions welcome; any member can answer any part.

Fields 1–3: title and abstract, research question, contributions

1 · Title and abstract

Working title: Fine-tuning BERTurk vs threshold-calibrated open LLMs for Turkish offensive language with 750 labels.

Small Turkish platforms can label only a few hundred tweets. We ask a narrow question: with 750 labelled OffensEval-TR tweets, does a fine-tuned BERTurk beat an open LLM that gets the same 750 labels as 8 prompt examples and 742 calibration tweets for its decision threshold? The TF-IDF + logistic-regression baseline scores 0.763 macro-F1 on the frozen 3,528-tweet test set, but only 0.620 with 750 labels. The LLM arm scores each tweet as log P(A) − log P(B), and the 742 labels set its threshold. On dev, calibration moves Qwen3-8B from 0.162 to 0.628 and Gemma-2-9B from 0.412 to 0.500; Qwen3-14B reaches 0.773 (zero-shot 0.171) and Gemma-4-12B 0.790 ± 0.007 (zero-shot 0.763). Fine-tuned BERTurk reaches 0.767 ± 0.014 on dev with 750 labels. **Gap:** the frozen-test comparison is pending; the final result uses the paired bootstrap.

2 · Research question

Unchanged. With 750 labels, does fine-tuning BERTurk beat prompting Gemma-2-9B-it or Qwen3-8B — same 750 tweets, 8 examples plus 742 threshold labels — on macro-F1 over the official 3,528-tweet test set?

"No" result: BERTurk's mean over 10 runs is not ≥0.02 above the LLM's mean over 3 runs, or the paired bootstrap says chance ($p \geq 0.05$). **Change:** none.

3 · Contributions

C1. At 750 labels, we test whether BERTurk's mean test macro-F1 is ≥0.02 above each LLM's, and whether a paired bootstrap rejects chance. Test pending; on dev BERTurk 0.767 vs 0.628 / 0.500.

C2. The 742-label threshold improves each LLM's dev macro-F1 over zero-shot; we report per-model delta and spread.

C3. The learning curve tests where fine-tuning overtakes TF-IDF between 250 and 6,000 labels; if the lines never cross, the 750-label answer stands.

Fields 4–5: related work and data

4 · Related work

1. **Zampieri et al. 2020**, SemEval-2020 Task 12. Top Turkish systems fine-tune on full data; no label-budget comparison.
2. **Çöltekin 2020**, LREC. The dataset; no fine-tuning-vs-prompting comparison.
3. **Zampieri et al. 2023**, NLE. Zero-shot LLMs; Turkish only Flan-T5-large 0.451; no examples.
4. **Dehghan & Yanıkoğlu 2024**, CASE. ChatGPT with 7 examples vs BERTurk, one dataset and one size.
5. **Pecher, Srba & Bielikova 2025**, EMNLP. English break-even near 100 labels; we test noisy Turkish two-class data.
6. **Schmitt & Schweter 2025**, SindBERT. Turkish encoders, full data only; no LLMs or small budgets.
7. **Şeker et al. 2025**, HATECAT-TR. Turkish span hate; BERTurk 0.59 vs LLaMA-3.3-70B 0.53; we classify whole tweets.
8. **Karakaş & Şimşek 2026**. Zero-shot LLMs vs BERTurk at one size; we add examples, calibration and a curve.

5 · Data

OffensEval-TR 2020, Çöltekin, LREC 2020; annotations CC BY 2.0; Turkish; @USER/URL masked. No extra cleaning.

Split	Tweets	OFF	OFF %
Official train	31,756	6,131	19.3
Our train	30,000	5,792	19.3
Dev	1,756	339	19.3
Test (frozen)	3,528	716	20.3

9 duplicate train texts; 1 test/train overlap. Example (OFF, id 33518): “Kırmızı ağır karar Allah belanı versin”.

Fields 6–9: method, protocol, results so far, error analysis

6 · Method and baseline

TF-IDF word 1–2 + char_wb 2–5, min_df=2, sublinear, Turkish lowercasing; LR C=1.0, balanced. Test 0.763 [0.745, 0.779].

Fine-tuning: BERTurk/XLM-R, 128 tokens, batch 16, 2e-5, 4 epochs or 400 updates, class-weighted, no early stopping. LLM: Qwen/Gemma 4-bit, 8 examples, 742-label threshold. One change: labels train weights vs choose examples and threshold.

Compute: about 3 Colab units for the 46 fine-tuning runs + 2.5 L4-h for 16 LLM passes.

7 · Evaluation protocol

Macro-F1; frozen 3,528-tweet test; 10 BERTurk runs at 250/750/2,000 and 3 at 6,000/30,000; 3 runs per LLM at 750. Paired bootstrap, 10,000 resamples; real gap ≥0.02 and wins on ≥95%. Fixed before method results; dev for code checks only.

8 · Results so far (mean ± sd) — BERTurk-30k = 0.832 ± 0.003

System	Dev	Test
Always NOT	0.447	0.444
TF-IDF, 750	0.613 ± 0.015	0.620 ± 0.012
TF-IDF, 30,000	0.763	0.763
Qwen3-8B zero	0.162	—
Qwen3-8B cal	0.628 ± 0.034	—
Qwen3-14B zero	0.171	—
Qwen3-14B cal	0.773 ± 0.009	—

System	Dev	Test
Gemma-2-9B zero	0.412	—
Gemma-2-9B cal	0.500 ± 0.048	—
Gemma-4-12B zero	0.763	—
Gemma-4-12B cal	0.790 ± 0.007	—
BERTurk, 250	0.680 ± 0.052	—
BERTurk, 750	0.767 ± 0.014	—
BERTurk, 2,000	0.793 ± 0.006	—
BERTurk, 6,000	0.806 ± 0.004	—
BERTurk, 30,000	0.832 ± 0.003	—
XLM-R, 750	0.670 ± 0.025	—

9 · Error analysis — 60 audited of 275

False alarms: no insult 9; political/religious 10; sports 5; personal 3; sarcasm 1; label noise 2. Misses: implicit insult 8; masked profanity 5; veiled threat 3; political/religious 4; institutional/sports 3; label noise 7. One example per category is in results/errors/audit.json.

Fields 10–12: research lead, plan and risk, reproducibility

10 · Research lead section — Vesile Han (CSE 654), fine-tuning; Gürkan Göztepeli (CSE 654), prompting

Vesile Han. Pecher et al., EMNLP 2025: English break-even near 100 labels, more for two-class tasks and variance; we test Turkish noisy two-class. Schmitt & Schweter, SindBERT 2025: Turkish encoders, full data only; no LLMs. Ablation A (done): BERTurk 0.767 vs XLM-R 0.670, +0.097 [0.080, 0.116], 100% wins. KUISAIL published BERTurk 0.816 on test; our 30k run should reproduce.

Gürkan Göztepeli. Şeker et al., HATECAT-TR 2025: BERTurk 0.59 vs LLaMA-3.3-70B 0.53 on Turkish hate spans; we classify whole tweets and score probabilities. Karakaş & Şimşek 2026: zero-shot LLMs vs BERTurk at one size; we add examples, calibration, curve. Ablation B: Gemma English instructions, 3 runs.

11 · Plan and risk

Done: BERTurk curve (36 runs), Ablation A (10). 21–24 Nov: Ablation B (Gürkan). 25–27 Nov: contamination check (Gürkan). 28–30 Nov: freeze, open test, final runs (all). 1–5 Dec: bootstrap, error analysis, paper (Mert · Utku). 8 Dec: report (all).

Biggest risk: 26B+ may not fit L4; Gemma-4-26B OOM'd at 22 GB, so 12B used. Fallback: Qwen3-14B alone. **Result risk:** lines may never cross; 750-label answer stands. **Proposal feedback:** add a test-set baseline; it is reported once, while methods stay on dev.

12 · Reproducibility

Repository: github.com/paysis/offenseval-tr-midproject. One command: `bash scripts/reproduce.sh gpu`. Seeds: split 42, subsets 42+k, bootstrap 42. Versions: Python 3.13.16, scikit-learn 1.9.1, transformers 5.22.1, torch 2.14.1, bitsandbytes 4-bit NF4.

Roles: to be confirmed before the final report. **AI tools:** a coding assistant was used for code, analysis and drafting; every number is read from saved result files and checked by the team.